GRACIAS POR INVITAR, LIVING PROJECT PORTAL, SPRINT 2

Estimation Appendix

How the team is sizing Release 1: the methods, the ranges and the reasoning. Part 1 takes the outside view, from three similar products; Part 2 estimates the event-goer slice two ways and presents a provisional range of 394 to 852 hours.

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Web	https://gpi-portal.onrender.com/sprint-2/estimation-appendix/

What this appendix is

The course asks for the estimation methods used, the ranges produced and the reasoning behind them. This page records them in the order the team produced them. Part 1 is the team's completed worksheet, "A First Look at Your Project's Size", which takes the outside view: the team's own experience and three similar products, as comparators for analogous estimation. Part 2 is the team's completed activity "Estimate One Slice of Your Project": one slice of the product, estimated by function points and by analogy, with story points as a cross-check, and the range a sponsor reads.

The entries are carried here as the team wrote them. In Part 1, two columns were tightened for the page: the evidence column cites the records in the interview evidence that each capability rests on, and each date in the comparators is matched to the listed source that states it, with the gaps marked. Part 2 is the team's submitted version of 30 September 2026, carried as submitted; the revision history below lists how it got there. The worksheet and the activity as delivered are the Team's PDFs above.

Part 1: a first look at the project's size

The outside view, before any counting: what Release 1 must do, the closest thing anyone on the team has built, and three real products that solve a problem close to GPI's.

Step 1: what Release 1 must do

Written as what the system must do, not how, from the charter's in-scope items and the strongest needs from the interviews. The hidden-work key: T, testing; S, security or privacy; I, integration with other systems; D, data (collection, cleaning, migration); H, hosting and deployment; U, usability and accessibility; O, documentation or training.

#	CAPABILITY (WHAT, NOT HOW)	EVIDENCE (CHARTER OR INTERVIEW)	HIDDEN WORK
1	An event organizer can upload events on the online platform	Project Charter, objective 1 (centralize events) and the in-scope calendar; P2-C3-10	U, O, D
2	An event organizer can pay for featured events on the online platform by day and time.	Project Charter, objective 2 (monetize on event publications) and the in-scope monetization of published events	U, O, D, H
3	Show a unified event calendar	Interviews (Phase 2): P2-C3-01, P2-C3-07, P2-C3-10, P2-C3-12; Project Charter, the in-scope centralized event calendar	T, D, H, U
4	Events goers will be able to subscribe to events based on categories	Interviews: P2-C3-11, P2-C3-05	D, U, I
5	The system will show a personalized feed that shows events based on the user's interests	Project Charter, the in-scope feed personalized by interests; interviews P2-C3-06, P2-C3-11, P2-C3-03, P1-C3-02	T, D, S
6	The system will show an event feed listed by date	Project Charter, the in-scope event feed listed by date; interview P2-C3-04	D, U, T
7	A user can save events they want to attend	Interview P2-C3-02	S, D, U
8	An administrator can review and approve a submitted event before it appears publicly	Project Charter, constraints: uneven data availability, and event planners	S, D, T, O

Step 2: the outside view, the team's own experience

The most similar thing anyone on the team has built before.

QUESTION	ANSWER
What was it?	KASAPro, the internal operations app Samuel built for KASA Contractors: job scheduling, route planning, and pricing. It is the closest match anyone on the team has built, a database-backed web app whose usefulness depends entirely on its records staying accurate and current, which is the same bet GPI is making on event data.

QUESTION	ANSWER
How long was it expected to take?	One month
How long did it take?	Three months
What took longer than expected?	The pricing and routing comparison: the dual unload-site cost comparison with load-type filtering, which carries more rules than anything else in the app. Every new load type changed the comparison, so the logic kept being reworked after it was supposedly done.

Step 3: the outside view, similar products

Three real products that solve a problem close to GPI's, at least one small and at least one big, measured from the start of development to the first public release. Under each entry, a source check notes which of the listed sources states each figure.

Product A, small: Upcoming.org

FIELD	ENTRY
Product and company	Upcoming.org, a community event calendar started by Andy Baio as a side project. Bought by Yahoo! in 2005.
Why it is similar to GPI	A city-by-city listing of what is happening, so people hear about events before they happen. The same job as GPI, built by one person with no budget.
Development started	January 2003 (notes from Baio's talk "A Tale of Three .Orgs," PDX Web Innovators, 4 June 2008; no Baio-written source for the start date, so treat it as secondhand)
First public release	19 September 2003 (Baio's own launch post on waxy.org)
Time to first release	About 9 months, worked nights and mornings around a full-time job. Baio's own verdict: "Rough site launched. Barely functional."
What the first release included	Anyone could add events they were going to, in fixed categories; see who else was going; browse by metro area; a friends list; RSS feeds. No API, tagging, email or SMS until March 2005.
Team size	One at launch. Baio kept his day job until the acquisition; two partners joined later.
What worked	Supply came from attendees posting what they were going to, not from promoters. Self-promotion was "banned entirely for its first two years." A city did not need broad coverage to be useful: "only 2 or 3 individuals" were enough to make a college town work.

FIELD	ENTRY
What did not work	After the founders left Yahoo! at the end of 2007, "the social features that made Upcoming unique were minimized, or removed entirely," and "spam, like creeping kudzu, was left unchecked." By 2009 "the only people using Upcoming were event promoters and spammers." A Kickstarter-funded rebuild funded 30 May 2014 did not relaunch until March 2017, about 34 months.
Sources	waxy.org/2003/09/upcomingorg_lau/; waxy.org/2013/04/the_death_of_upcomingorg/; stubborndreams.me/2008/07/31/a-brief-history-of-upcoming-dot-org-by-andy-baio/; en.wikipedia.org/wiki/Upcoming

Source check. The January 2003 start, the nine months, the nights and mornings around a day job, "only 2 or 3 individuals" and the October 2005 acquisition are in the 2008 write-up of Baio's talk. The 19 September 2003 launch is Baio's own launch post, confirmed by Wikipedia. The end-of-2007 departure, the minimized social features, the spam and the 2009 line are in Baio's 2013 post. Wikipedia dates the Kickstarter campaign's launch to 7 May 2014. Not stated in any listed source: the "banned entirely for its first two years" quote, "Rough site launched. Barely functional.", the March 2005 feature dates, the 30 May 2014 funding date and the March 2017 relaunch.

Product B, big: Do512

FIELD	ENTRY
Product and company	Do512.com is the Austin event discovery platform created by the DoStuff media network, run by local teams in 20 cities.
Why it is similar to GPI	It serves as a discovery platform that helps people stay up to date with local events and places, which is what GPI will try to achieve.
Development started	It started in Austin, TX, in 2006. ("Meet Courtney Goforth of Do512", Voyage Houston)
First public release	2006 ("The Story", Do512.com). They founded DoStuff in 2010, which let them expand to more cities. ("Innovation Nation: DoStuff Media", Austin Monthly)
Time to first release	Do512's first release took 1 month after they first came up with the idea. ("Innovation Nation: DoStuff Media", Austin Monthly)
What the first release included	It started as a website that listed all happy hours around Austin, but it then expanded to include all sorts of events. ("The Story", Do512.com)
Team size	Two people at first (Jimmy Stewart and Scott Owens). Soon after, they recruited their girlfriends, so it became a team of 4.

FIELD	ENTRY
What worked	Broadening the scope of the website (including all types of events, not just happy hours). Dynamic listings. Adding more people to the team. External users (venues, artists, event organizers) could upload and update their own events, that ensured the list was accurate and exhaustive. Partnered with local venues to offer free tickets to users who would rsvp to their events, creating a lot of engagement. Creating their media DoStuff label to be able to use the same technology in other cities.
What did not work	Having a smaller team of people. Having the scope of the site be so narrow. Static website, because it led to inaccuracy and outdated data. The first few years the business was not making money, it took them a few years to gain revenue.
Sources	crunchbase.com/organization/dostuff-media; voyagehouston.com/interview/meet-courtney-goforth-do512-austin-texas/; austinmonthly.com/innovation-nation-dostuff-media/; do512.com/p/do512s-10-year-anniversary-the-story

Source check. The Voyage Houston interview states the 2006 start in Austin, the founders Jimmy Stewart and Scott Owens, and the happy-hours catalogue the site began as. The Austin Monthly article, Do512's own story page and Crunchbase refuse automated reading, so the one-month build, the team of four, the 2010 founding of DoStuff, the free-ticket partnerships and the revenue statement could not be checked from the portal; they stand on the worksheet's attribution to those pages.

Product C, big: Fever

FIELD	ENTRY
Product and company	Fever, founder by 4 individuals, is a personalized event discover web platform
Why it is similar to GPI	Initially started as a event discovery for two cities, quickly grew into hundredths of cities/categories/events, matching out interest-based feed.
Development started	2012, New York, NY
First public release	Released in 2014 simultaneously in New York and Madrid
Time to first release	About 2 years.
What the first release included	Personalized recommendations for entertainment, dining, and cultural events, live in two cities.
Team size	4 co-founders at launch
What worked	Launching in two major cities (New York & Madrid)

FIELD	ENTRY
What did not work	Not documentation in the sources found.
Sources	feverup.com/en/mexico-city; newsroom.feverup.com/en-AU/about/; preqin.com/data/profile/asset/fever-labs--inc-/140163

Source check. The Preqin profile states that Fever Labs was founded in 2014, is headquartered in New York, and has four founders (Pep Gomez, Alexandre Perez Casares, Francisco Hein and Ignacio Bachiller Strohlein). The two Fever pages state no dates, founders, cities or team size. Not stated in any listed source: the 2012 start, the Madrid launch, the two years to first release, and what the first release included.

Step 4: what the comparison says

1. **Which hidden-work letter appears most often, and did the similar products struggle with it?** Data (D) appears the most often, which can be seen in all 8 capabilities. Do512 also struggled with static data which led to inaccurate and outdated listings. Upcoming.org also struggled with data quality and moderation, eventually dying out when unmoderated spam ruined the listings.
2. **How long did the similar products take, with how many people, and what does that suggest?** Upcoming.org took roughly 9 months for a first release, with only one person. Do512.com took 1 month for the first release; initially it was two people, they then recruited two more people. Fever took about two years for the first release; there were four co-founders for the first release. Although we deployed a functional baseline website in one week after doing research, our comparables indicate that developing a production-ready first release would take about 6 to 12 months.
3. **One practice to copy and one mistake to avoid.** Practice to copy: allow external users and organizers to directly submit and manage event content. Mistake to avoid: avoid static listings since it leads to inaccuracy and outdated data.

Our comparables suggest a first release would take 6 to 12 months with a team of 5.

Part 2: one slice, estimated two ways

The team's completed activity "Estimate One Slice of Your Project" (Lesson 9, Part 2), carried here as the team submitted it on 30 September 2026. One slice of the product is estimated twice, with two methods that start from different information; the two answers are compared, and a range is presented. The event supply follows the charter and the strategy: organizers post, the team curates at first, and no outside-source import is in the slice, so both methods estimate the same functions.

The figures stay provisional on the rates: the function-point productivity rate and the analogous inputs are assumed, not measured. The team confirmed the seven story point values and verified the wage figure on 29 September 2026.

Step 1: the slice

QUESTION	OUR SLICE
Slice name (one or two major parts)	User and event-goer features
Why we chose it	Event discovery is the central user experience. This coherent slice covers the calendar, date views, category preferences, saved events and search; organizer and admin flows are separate.
Screens or forms where users enter data	Log in and user registration; the account settings page; the category ("interest") selection screen; the save event button
Reports, results or lookups users see	Outputs: a feed ranked by the user's chosen categories. Personalization in this slice filters or orders events by user-selected categories; machine-learning and predictive recommendations are not included. Lookups: the calendar of El Paso and Juárez events, and the same events listed by date (one lookup, two presentations); the event details page (time, place, details); search results from the search bar; the saved events list; my account and chosen categories.
Data the slice stores	Users; the user's preferences; events
Other systems it connects to	None in this slice. Source of events: this slice reads a common GPI events catalogue. For Release 1, the Project Charter and the Business Strategy have organizers posting events, with the team curating at first, and this estimate takes that as the supply.
Out of the slice	Organizers uploading or editing events; importing events from outside sources; paid featured listings for organizers; admin review and approval of submitted events; ticketing and payment processing (out of scope in the charter); email or push notifications about new events; social features, like a friends list or seeing who else is going

Stories in the slice.

USER STORY	MOSCOW	POINTS
As an event goer, I can see a unified calendar of events across El Paso and Juárez.	Must	8

USER STORY	MOSCOW	POINTS
As an event goer, I can browse upcoming events listed by date so I know what is on this weekend.	Must	5
As an event goer, I can open an event and see its time, place and details.	Must	3
As an event goer, I can create an account and subscribe to the event categories I care about.	Must	8
As an event goer, I can see a feed ranked to the categories I chose.	Should	5
As an event goer, I can save an event so I can come back to it later.	Could	3
As an event goer, I can search for an event by name or keyword.	Should	5
Total points		37

Points: on 29 September 2026 at 3:30 PM, the team reviewed and confirmed all seven user-story point values. The agreed values are 8, 5, 3, 8, 5, 3 and 5, totaling 37 story points. Search was added to the story table in an earlier revision after appearing among the slice lookups without a separate story. Capabilities 5 and 7 of the Part 1 list, the personalized feed and saved events, are Should and Could here: both are in scope for Release 1, and the Business Strategy sequences personalization after the core listing.

Step 2: the two methods

Analogous and function points. In the team's words: "Analogous method begins at the market research level where we identify facts such as failures and successes of a product. We can prove through research and facts about how an organization might have shipped a similar product and succeeded or failed. Function points, on the other hand, start from one's own design indicating count of inputs, results, lookups and data in our slices. Neither method uses the other's numbers, so if they agree, the agreement means something." Use-case points were not used. Story points (3C) were completed as a cross-check, not as a declared method.

Step 3A: function points

CATEGORY	COUNT	WEIGHT	POINTS
External inputs (EI)	4	4	16
External outputs (EO)	1	5	5
External inquiries (EQ)	5	4	20

CATEGORY	COUNT	WEIGHT	POINTS
Internal logical files (ILF)	3	10	30
External interface files (EIF)	0	7	0
Total function points			71

Rate: low 6 and high 12 hours per function point, the range from Part 2, Slide 15, since the team has no productivity history (assumption). Effort: 71 × 6 = 426 hours (low); 71 × 12 = 852 hours (high).

The counts, itemized. External inputs (4): user registration, account settings update, category interest selection, and save event. Log in is listed as a screen in Step 1 but is not counted as an input, because it stores no business data of its own. External output (1): the category-ranked feed, which orders events against stored preferences and so derives data rather than returning it as stored, which is what separates an output from an inquiry on Slide 13. External inquiries (5): events listed by date, which the calendar presents in another form over the same data and processing and so is not counted again; the event detail page; search results; the saved events list; and the account and chosen categories view. Internal logical files (3): users, user preferences, and events. External interface files (0): the slice reads only the GPI catalogue; the outside source carried in the previous revision is out of the slice. Every item appears once and none is counted in two categories.

Step 3C: story points, as a cross-check

INPUT	VALUE	NOTE
Slice points	37	From the Step 1 story table
Team size	5	Five or six at most
Hours per person per sprint	20	Assumption
Velocity: low, middle, high	4, 6, 8	Assumptions until there is sprint data
Sprints needed (high to low velocity)	4.6, 6.2, 9.3	Slice points ÷ velocity
Effort in hours (low to high)	463 to 925	Sprints × team size × hours per person per sprint

Our assumptions: twenty hours per person per sprint is a two-week sprint at 10 productive hours a week each, alongside coursework. The velocities of 4, 6 and 8 points per sprint have nothing behind them: Slide 26 says real velocity needs at least three completed sprints by the same team, and we have run none. Effort is sprints × 5 people × 20 hours, so 100 team hours a sprint, giving 463

hours at the high velocity, 617 at the middle and 925 at the low. Low effort corresponds to the highest velocity.

Step 3D: analogous

QUESTION	ANSWER
Comparable product, team size, and time to first release	Upcoming.org. One person (Andy Baio), about 9 months to first public release, 19 September 2003 (waxy.org launch post).
Person-months (team size × months)	1 × 9 = 9 person-months. Baio built it nights and mornings around a full-time job, so we assume 70 hours per person-month rather than 160 (assumption). That puts their first release at roughly 630 hours.
Fraction of their release our slice matches (low to high), and why	50 to 70 percent. Their first release was browse-by-metro listings in fixed categories, user-added events, attendee lists, a friends list and RSS. Our slice takes the browsing and listing side: their metro browse against our calendar and date list, their event page against ours, and their events-you-are-going-to list against our saved events. We add account preferences and a category-ranked feed, which their first release did not have. Later updates added an API, tagging, email and SMS (the sources we listed do not date them), and we found no source showing it ever carried a ranked feed, so we do not claim one. We leave out their event entry, attendee lists and friends list, which sit outside our boundary. Both methods now cover the same functions: our slice reads events from the GPI catalogue, as Upcoming's first release read the events its users had added, and the entry of events sits outside our boundary in both.
Team size adjustment and why	Uplift of 25 to 35 percent. Upcoming was one person with no coordination cost at all. We are five, with 10 communication paths, so our effort per feature is higher than his. We did not scale from Fever's four founders, whose first release took about two years by our Part 1 worksheet (a figure the listed sources do not confirm), because that was a company rather than a side project and its effort already carries its own coordination.
Adjusted effort in hours (low to high)	630 × 0.50 to 0.70 = 315 to 441 hours. With the 1.25 to 1.35 uplift, 394 to 595 hours. Middle, at 60 percent and 1.30, is 491 hours.
Confidence, and why	Medium. The comparable is the right shape, and the 19 September 2003 first release is grounded in Baio's own launch post on waxy.org. The January 2003 development start is secondhand, from one attendee's notes on a talk he gave in 2008, and we found nothing written by him giving that date. Beyond the dates, the 70 hours per person-month, the 50 to 70 percent fraction and the 25 to 35 percent uplift are all our modeling choices rather than recorded figures. The fraction moves the answer by about 40 percent across its range and the uplift by about 8 percent; the hours per person-month move it without limit: at 100 hours per person-month the range would be 563 to 851 hours, and at a full-time 160, 900 to 1,361.

Step 4: the team-size check

CHECK	YES OR NO	WHY IT MATTERS
Our team is five or six people at most.	Yes, a team of 5	Beyond six, split into two teams.
We counted our communication paths: n(n − 1) ÷ 2.	Yes, 10 paths	Five people share 10 paths; ten share 45.
We added explicit work for integrating the pieces of the slice.	No; see the integration work below.	Breaking work into pieces hides the cost of putting it back together.
We did not assume that more people means proportionally more output.	Yes	Coordination grows faster than headcount.
We estimated for the team we have.	Yes	Not for a larger team we wish we had.
We adjusted, not copied, timelines from much larger teams.	Not applicable. Our comparable was one person, smaller than our team, so we scaled its effort up (3D) rather than copying a larger team's timeline down.	Their effort includes their coordination costs.

Integration work (row 3). The answer is No because no separate hours were added for putting the pieces together. Coordination and technical integration are different things: the 25 to 35 percent uplift in the analogous estimate covers coordination among five people, and the 6 to 12 hour function point rate is an all-in rate from design through testing. The lines below are assumed inside that rate and inside the comparable's recorded effort, which included its own integration.

- Keeping organizer-entered dates, times, places and categories consistent across the two cities: assumed inside the function point rate.
- Wiring the calendar, detail page, ranked feed and saved events together: assumed inside the function point rate.
- End-to-end tests, data-failure handling and deployment: assumed inside the function point rate.

Step 5: the comparison

	FUNCTION POINTS	ANALOGOUS
Estimate in hours (low to high)	426 to 852	394 to 595
Middle value in hours	639 (71 FP × 9 h)	491
Input that drives it most	Productivity rate (hours per function point)	Hours per person-month assumed for the comparable (70), then the fraction of the release matched
Measured or assumed?	Assumed	Assumed

Gap: larger middle value ÷ smaller middle value = 639 ÷ 491 = 1.30, an absolute difference of 148 hours.

1. **Do the two methods depend on the same assumption?** No, the function points method starts from our internal design counts (inputs, reports, results, etc.) in Step 1, multiplied by productivity rates. The analogous method starts from external market historical data scaled to our scope and adjusted for 5-person team coordination costs.
2. **Which single input moves the answer most?** The productivity rate in function points moves the estimate by 426 hours, from 426 at 6 h/FP to 852 at 12. In the analogous method, holding the coordination uplift fixed at 30 percent and moving the matched-release fraction from 50 to 70 percent takes the estimate from 409.5 to 573.3 hours, a difference of 163.8 hours. The full 394 to 595 range varies the uplift as well, between 1.25 and 1.35, so that wider 201.6 hour spread cannot be attributed to the fraction alone. The analogous method's lever is the 70 hours per person-month assumed for Baio's side project: at 100 hours per person-month the range would be 563 to 851 hours, and at a full-time 160, 900 to 1,361 hours, more than the fraction or the uplift can move it.
3. **What information would narrow the gap, and when could you get it?** Our first one or two actual software-development sprints would give early effort observations, but they would not establish a reliable rate. We need hours recorded against completed stories, and Slide 26 sets at least three completed sprints by the same team before velocity means anything. Course-portal documentation sprints do not count as GPI development velocity. More data should narrow the 6 to 12 hour range; it will not eliminate the uncertainty.

The team's explanation of the gap. Function point yielded a middle value of 639 hours, whereas the Analogous method yielded a middle value of 491 hours, creating a gap of 1.30x (under 2.0x). The primary driver of the gap is that our Function point estimate assumes a wide productivity rate of 6 to 12 hours per point due to lack of sprint data. Meanwhile, the Analogous method relies on our assumed Fraction of their release our slice matches and a 70-hour per person month. Both methods

start from completely independent information, so the gap represents the distance between two assumptions we have never tested. The optional story-point cross-check gives 463 to 925 hours for the team's confirmed 37-point slice. Its result of 925 hours exceeds the two-method upper bound of 852 hours by 73 hours. Because the cross-check velocity is an untested assumption and story points are not a declared estimation method, this indicates upper-end uncertainty rather than automatically changing the formal range. What would narrow the gap is real hours logged against finished stories. Holding the coordination uplift at 30 percent and moving the feature match from 50 to 70 percent takes the analogous estimate from 409.5 to 573.3 hours, a swing of 163.8 hours. The wider 394 to 595 spread varies the uplift as well, so the fraction alone does not account for all of it.

Step 6: the estimate

FIELD	OUR ESTIMATE
Slice estimated	User and event-goer features
Range in hours (low to high)	394 to 852 hours
Expected hours: (O + 4M + P) ÷ 6, and why we chose M	(394 + 4 × 594 + 852) ÷ 6 = 603.67 expected hours. For M we took the midpoint of the lowest and highest ends of the range, which gave a first value M1 = 623, then the midpoint of the two methods' middle values, which gave M2 = 565. We settled on the midpoint of those two, M = 594. M = 594 is a planning judgment, not a measured value. We think an outcome near 594 hours is plausible because both methods rely on assumed data, and 594 sits in the middle, where neither method's weakness dominates.
Cost range and loaded rate used	Base hourly wage: $51.36, for ZipRecruiter's "IT Software Engineer Salary in Texas" listing (source). The team verified the figure and job category on 29 September 2026 at 3:30 PM. Overhead: 20 percent, a team assumption, producing a loaded hourly rate of $61.63 (rounded). Low: 394 × 61.63 ≈ $24,282. Expected: 603.67 × 61.63 ≈ $37,204. High: 852 × 61.63 ≈ $52,509.
Calendar estimate in weeks, and hours per week assumed	Expected: 12.07, about 12 weeks, at an assumed 10 productive hours per person per week. Low end: 394 ÷ 50 ≈ 7.9 weeks. High end: 852 ÷ 50 ≈ 17.0 weeks. This assumes all five people are available continuously at 10 productive hours a week. It is a production estimate for the slice. In Part 1, we estimated that it would take 6 to 12 months to produce a full first release; however, after Part 2 of the estimation analysis, 12.1 weeks corresponds to the time it would take to implement the event-goer slice alone, at the stated capacity.
Methods used	Analogous and function points. Story points (3C) were completed as a cross-check, not as a declared method.

FIELD	OUR ESTIMATE
Key assumptions	Productivity rate: 6 to 12 hours per function point, from Part 2, Slide 15; with no sprint history it is not measured. Scope: as listed in Step 1. 70 hours per person-month for Baio's side-project effort on Upcoming, rather than a full-time 160. Our slice is 50 to 70 percent of Upcoming's first release. A 25 to 35 percent coordination uplift for five people. 20 percent overhead on the ZipRecruiter hourly figure, a team assumption. 10 productive hours per person per week, which drives every week figure. Velocities of 4, 6 and 8 points per sprint in the story point cross-check. Event supply: organizer-posted events in a GPI-owned catalogue, with the team curating at first, as the Project Charter and the Business Strategy have it; no outside-source import is in this slice. The team reviewed and confirmed the seven story-point values (total 37) on 29 September 2026 at 3:30 PM. Story-point velocity (4, 6 and 8 points per sprint) remains an assumption because no completed software-development sprints have established historical velocity.
Team size assumed	5 team members
What would change this estimate	Adding more productive hours per person. Adding an outside-source import to the slice. Further feeds of the same data add integration hours, not function points. Real productivity and velocity data after three completed sprints, which Slide 26 sets as the minimum. Scope changes. Team changes.
Date of this estimate	29 September 2026

What is still open

- Measured rates. The productivity rate and a real velocity come only from completed development sprints, at least three by the same team (Slide 26); until then every rate on this page is an assumption.
- What this range feeds next. The Project Budget prices it. The sponsor confirmed on 5 October 2026 that a Business Case and an ROI Analysis are not Sprint 2 deliverables, so the range's next use comes with a later sprint. The change log records how the range moved and what it affected.

Delivered by the team: Project Estimation Starter, Part 1: the team's worksheet as completed (https://gpi-portal.onrender.com/docs/sprint-2-estimation-starter.pdf); Estimate One Slice of Your Project, Part 2: the team's activity as submitted, 30 September 2026 (https://gpi-portal.onrender.com/docs/sprint-2-estimation-activity.pdf).

Generated from the portal source at build time. The web page and this PDF are two renders of the same document. Revision history: v0.1 (Sep 24, 2026) Part 1, the team's completed worksheet 'A First Look at Your Project's Size', carried onto the page as written. The evidence column cites the interview records each capability rests on, and each comparator date is matched to the listed source that states it. Part 2, the two estimates, to follow.; v0.2 (Sep 29, 2026) Part 2, the team's completed activity 'Estimate One Slice of Your Project', carried onto the page: the slice, the two methods, the story-point cross-check, the team-size check, the comparison and the presented range. Edits from the team's alignment review of 29 September: the source-of-events and personalization boundaries stated in Step 1; the sensitivity of the analogous estimate corrected, with the hours per person-month shown as its largest lever; two claims the Part 1 source checks mark unsupported reworded; the comparable's missing import work noted; the larger-teams check answered not applicable; the PERT display rounded to 669.67; the open decision on event supply recorded with what it changes; the story points noted as drafted, not sized by planning poker. No count, weight, rate, fraction, uplift, velocity or range changed.; v0.3 (Sep 29, 2026) Items 1 to 4 of the team's final revision handoff. The outside-source import leaves the slice so both methods estimate the same functions, as the charter and the Business Strategy have the supply; the calendar counts as the same lookup as events by date, under the handoff's own rule; Steps 3A, 5 and 6 re-run from the team's unchanged counts, weights, rates, fraction, uplift and its documented procedure for M: 71 function points, 426 to 852 hours, a gap of 1.30, a range of 394 to 852 hours and 603.67 expected. The alternative-count sentence is now conditional with its arithmetic corrected, the story points are labelled unconfirmed pending the team's validation, and the wage listing carries its actual category. The slice boundary and the calendar classification are recorded as revisable by the team.; v0.4 (Sep 29, 2026) The team's verification of 29 September 2026, 3:30 PM, recorded in the team's own wording: the seven story point values reviewed and confirmed (8, 5, 3, 8, 5, 3 and 5, total 37), and the $51.36 base wage and its job category verified on the ZipRecruiter listing. The story-point cross-check is now compared to the formal range correctly (its 925-hour end exceeds the 852-hour upper bound by 73 hours, a sign of upper-end uncertainty, not a change to the range), and the figures stay provisional only on the assumed rates. No calculation changed.; v0.5 (Sep 30, 2026) The team's submitted version of the activity replaces version 9 as the Team's PDF: Part 2, and the page follows its final edits: the source-of-events note shortened, the Step 4 closing paragraph and the term-length caveat dropped, the rationale for M and the gap paragraph tightened, and the import line under 'What would change this estimate' reduced to one sentence. No figure changed.; v0.6 (Oct 5, 2026) The closing note no longer announces a Business Case and an ROI Analysis for this sprint: the sponsor confirmed on 5 October 2026 that Sprint 2 requires the two estimation activities and the budget. It now points to the Project Budget and to the change log. No other change.; v1.0 (Oct 5, 2026) Approved by the team as published, and its owner assigned: Christian Lopez-Matulessy, Research Documentation Lead, whom the team's budget names as the appendix owner. The label of the 0.5 revision, shown as 0.6 since the 0.6 revision was added, is corrected. No content change.